

GroupFunctions.jl: computing individual entries of the irreducible representations of the unitary group $U(d)$

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Summary

[GroupFunctions.jl](#)¹ is a Julia library for computing individual matrix elements of irreducible representations of $U(d)$. These matrix elements, called group functions, can be evaluated symbolically or numerically. For $SU(2)$, they reduce to the Wigner D -functions. The library computes these matrix elements in a carrier-space basis enumerated by Gelfand-Tsetlin patterns (Gelfand & Tsetlin, 1950). It can also compute entire representation operators, construct input unitaries from parameterisations common in quantum optics, translate Gelfand-Tsetlin patterns into occupation-number kets, and compute the associated Schur functions. Results can be exported in a form compatible with Mathematica.

Statement of need

Representations of the unitary group $U(d)$ arise in many subfields of physics and mathematics, and computations often reduce to evaluating their matrix elements, called group functions. For an irrep labelled by λ , the corresponding object is

$$D_{\text{out,init}}^{(\lambda)}(U) = \langle \text{out} | D^{(\lambda)}(U) | \text{init} \rangle,$$

where `init` and `out` denote basis states in the representation carrier space.

In mathematics, summing the diagonal group functions gives the trace of the representation matrix of U . This trace is the character of the representation and the Schur polynomial of the eigenvalues of U , an object central to algebraic combinatorics and symmetric function theory.

In quantum physics, group functions appear in several settings. In quantum optics, a group function gives the transition amplitude of photons through a linear optical network (de Guise et al., 2014). The same object can be used to characterise the performance of quantum devices (Amaro-Alcalá, 2026) and to describe the symmetry properties of states (Otto & Szymański, 2024). One important subproblem is boson sampling (Aaronson & Arkhipov, 2011), where the transition amplitude reduces to a permanent whose evaluation is classically computationally hard. Whether noisy real-world quantum devices can perform this computation remains an open question.

Some of these tasks require only a numerical estimate, whereas others require an exact symbolic group function. `GroupFunctions.jl` addresses the latter. Although related packages exist (see below), to our knowledge, none are designed to compute the representation matrix symbolically,

¹David Amaro-Alcalá developed the package, its algorithms, and its test suite, and prepared the initial documentation. Konrad Szymański substantially revised and expanded the documentation and developed additional examples, and contributed to the standardisation of the API.

one matrix element at a time – the cheaper route when a calculation requires only a few of them.

State of the field

Several existing packages relate to `GroupFunctions.jl` but address different problems. `SUNRepresentations.jl` ([QuantumKitHub contributors, 2020](#)) and the algorithm presented in the appendix of ([Alex et al., 2011](#)) compute $SU(d)$ Clebsch-Gordan coefficients. `RepLAB` ([Rosset et al., 2021](#)) supports manipulating irreducible representations of various groups, including $U(d)$, but provides only indirect numerical access to group functions. `IntegrateUnitary.jl` ([Pawela & Puchała, 2026](#)) performs symbolic integration over compact groups rather than evaluating representation matrices. `haarpy` ([Cardin et al., 2024](#)) implements Weingarten-calculus methods. Other libraries focus on quantum optics. `BosonSampling.jl` ([Seron & Restivo, 2024](#)), `Perceval` ([Heurtel et al., 2023](#)), and `Q0ptCraft` ([Aguado et al., 2023](#)) numerically model linear optical devices, while `The Walrus` ([Gupt et al., 2019](#)) helps compute amplitudes for Gaussian boson sampling. These packages do not target the symbolic computation of individual representation-matrix entries, which is the primary purpose of `GroupFunctions.jl`.

Software design

This library provides a unified method for computing representation matrix elements of $U(d)$ irreps specified by integer partitions of length at most d , including computations with symbolic input matrices. Several computational routes are possible in principle. For symmetric irreps, which model fully indistinguishable bosons, one can manipulate states as polynomials of creation operators applied to the vacuum. Another approach constructs and exponentiates the Lie algebra generators in the chosen representation. Both approaches are computationally expensive. More restricted methods based on generating functions also exist ([Prakash, 1996](#)).

The authors chose the Grabmeier-Kerber formula ([Grabmeier & Kerber, 1987](#)) as the most general solution. It expresses the matrix element as a sum of monomials in the entries of the input matrix, weighted by the irreducible representation and the states in question. Our implementation optimises the enumeration of the double cosets that index this sum by grouping permutations that contribute the same monomial. This implementation provides the library's main function, `group_function`.

Internally, the algorithm represents basis states as semistandard Young tableaux. The user-facing functions expose the equivalent Gelfand-Tsetlin patterns through the `GTPattern` data structure and provide utility functions for common quantum optics scenarios, such as `occupation_number`.

Research impact statement

`GroupFunctions.jl` has been used to evaluate matrix elements and group characters in published work on filtered randomized benchmarking ([Amaro-Alcalá, 2026](#)). The library has been under continuous development since 2020 and is tested in CI; it is registered in the Julia General registry under the MIT licence, installable with the following command:

```
] add GroupFunctions
```

The following example evaluates, symbolically, a matrix element between states of the $U(4)$ mixed-symmetry irrep.

```
λ = [2,2,1,0]; basis=basis_states(λ); # integer partition and basis
group_function(λ, basis[1], basis[end]) # symbolic matrix element
```

The result is a polynomial in the entries of the $U(4)$ matrix. For symmetric irreps the matrix element reduces to a permanent, which symbolic algebra packages also compute; for mixed-symmetry ones no existing software computes these elements symbolically. Further examples, applications, and mathematical background are available in [the documentation](#).

AI usage disclosure

OpenAI Codex (GPT-5.3) assisted with optimising the performance of the double-coset enumeration at a late stage. The authors developed the mathematical design and proof of correctness of the optimised algorithm. Anthropic Claude (Opus 4.8) assisted with code review and language review of the documentation and manuscript. The authors reviewed and edited all AI-assisted changes.

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